Software Test Plan (STP)

for

Student Attendance Management System (SAMS)

Version: 1.2

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1. Introduction

This document defines the test plan for the Student Attendance Management System (SAMS) version 1.0. It gives the objectives, scope, approach, environment, schedule, responsibilities and test cases for testing SAMS against the requirements in the SRS.

Objectives:

- Verify all functional requirements (REQ-1 to REQ-19)
- Validate performance and security requirements (NFR-1 to NFR-11)

- Make sure every SRS requirement is covered by atleast one test case

References:

- SAMS SRS v1.0
- SAMS SAD v1.0
- IEEE 829 Standard for Software Test Documentation

2. Test Items

Module (Django app)	**What it does**
accounts	Login, logout, roles, user management, Excel upload
academics	Subjects, classes, enrolment, faculty assignment
attendance	Mark / update attendance, percentage, change log
reports	Student view, PDF / Excel reports, shortage list
leaves	Leave requests, approval, shortage email

Build under test: SAMS v1.0 from the main branch of the team GitHub repository.

3. Features to be Tested

Req ID	**Feature**
REQ-1, REQ-2	Login and role based dashboard
REQ-3, REQ-4	Add / edit / delete users, Excel upload
REQ-5 to REQ-7	Subjects, classes, faculty assignment
REQ-8, REQ-9	Mark attendance, no duplicate marking
REQ-10	Update attendance within 3 days
REQ-11	Automatic percentage calculation
REQ-12	Student views own attendance
REQ-13, REQ-14	Report generation, PDF / Excel download
REQ-15	Shortage list
REQ-16	Shortage email

REQ-17 to REQ-19	Leave apply, approve / reject, leave not counted absent
NFR-1 to NFR-3	Performance
NFR-6 to NFR-11	Security

4. Features Not to be Tested

- Internal working of Django, MySQL, reportlab and openpyxl (assumed tested by thier developers)
- Gmail delivery after the mail is handed to the SMTP server
- Load above 100 users (outside SRS scope)
- Other languages (only English supported)

5. Test Approach / Strategy

Level	**What**	**Tool**
Unit testing	Models, views and helper functions of each app	Django TestCase
Integration testing	Flows across apps, e.g. mark attendance -> shortage email	Django test client
System testing	Full end to end flows in the browser	Manual (browser)
Acceptance testing	Demo scenarios with the faculty	Manual

Types of testing:

- Functional testing of every REQ
- Boundary testing - 74.99% vs 75%, 3 day edit limit
- Negative testing - wrong inputs, unauthorised access
- Performance testing - page load and report time
- Regression testing - rerun all unit tests before every merge to main

Entry criteria:

- Code merged to main and running on the test machine
- Test data loaded

Exit criteria:

- All High priority test cases executed
- Atleast 95% test cases passed
- No open critical defects

5.1 Security Validation

Security testing is done against the security objectives SO-1 to SO-3 and requirements NFR-6 to NFR-11 in the SRS.

Security Check	How it is tested	Req
Password hashing	Check user table in MySQL - password should not be plain text	NFR-6
Role based access	Login as each role and try to open pages of other roles - should get 403	NFR-7
Student data privacy	Student changes URL to view another student's attendance - should get 403	NFR-8
Session timeout	Stay idle for 31 minutes - should be logged out	NFR-9
CSRF protection	Submit a form without CSRF token - should be rejected	NFR-10
SQL injection	Enter ' OR 1=1 -- in login and search fields - should not work	NFR-10
Audit log	Edit an attendance record - log entry should be created with name and time	NFR-11
Sensitive data in logs	Check log files - no passwords or phone numbers	SO-3

6. Test Environment

Item	Details
Hardware	Team laptops (MacBook Air / Windows laptops, 8 GB RAM), Android phone for mobile view
OS	macOS 14, Windows 11
Software	Python 3.10, Django 4.2, MySQL 8.0

Browsers	Chrome, Firefox, Safari (latest)
Test data	2 classes, 5 subjects, 70 students, 4 faculty, 1 admin

7. Test Schedule

Milestone	**Date**
Test plan and test case design	05-10-2026
Test environment setup	12-10-2026
Unit and integration testing (Sprint 1)	19-10-2026 to 23-10-2026
Unit and integration testing (Sprint 2)	26-10-2026 to 30-10-2026
System and security testing	02-11-2026 to 06-11-2026
Regression and acceptance testing	06-11-2026
Test reports (unit, system validation, code and branch coverage)	06-11-2026

8. Test Deliverables

- Test Plan (this document)
- Test cases
- Unit test scripts (tests.py in each app)
- Test data
- Defect list (JIRA bugs)
- Test summary report

9. Roles and Responsibilities

Role	**Name**	**Responsibility**
QA Lead	Epari Subhransi	Prepare test plan, coordinate testing, report results, test reports module
Test Engineer	Cheethirala Sai Santosh	Test accounts module and security checks
Test Engineer	Deepak B	Test academics and attendance modules,

		performance tests
Test Engineer	Gyaan S Shetty	Test leaves module and shortage email, maintain test environment

10. Risks and Mitigation

Risk	**Mitigation**
Build not ready on time	Test each module on its own branch first
Test machine / MySQL not working	Each member has local MySQL setup
Email cannot be checked	Use Django console email backend while testing
Less time at end of semester	Write unit tests along with code in every sprint

11. Assumptions & Dependencies

- MySQL and Django are installed on all team laptops
- Test data represents a real class (upto 70 students)
- A Gmail account is available for email testing
- Requirements in SRS v1.0 will not change during testing

12. Suspension & Resumption Criteria

Suspend testing if:

- Login or attendance marking is not working (blocks other tests)
- A critical security defect is found
- Test environment is down for more than 4 hours

Resume when:

- The blocking defect is fixed and retested
- Environment is back up

13. Test Case Management & Traceability

Test cases are given below. Each test case is mapped to SRS requirements, and the traceability table shows that every requirement is covered.

Test Cases

TC ID	Req ID	Test Description	Precondition	Expected Result	Priority	Result
TC-01	REQ-1	Login with valid ID and password	User exists	Dashboard opens	High	Pass
TC-02	REQ-1	Login with wrong password	User exists	"Invalid username or password" shown	High	Pass
TC-03	REQ-2, NFR-7	Student tries to open admin page	Logged in as student	403 Forbidden shown	High	Pass
TC-04	REQ-4	Upload Excel with 10 students (2 with missing SRN)	Logged in as admin	8 students added, 2 rows rejected with reason	Medium	Pass
TC-05	REQ-5, REQ-6	Add subject and class, assign students	Logged in as admin	Subject and class saved	High	Pass
TC-06	REQ-8	Mark attendance with 3 absentees for class of 60	Faculty assigned to subject	57 Present, 3 Absent saved	High	Pass
TC-07	REQ-9	Mark attendance again for same subject, date and hour	TC-06 done	Error "Attendance already marked"	High	Pass
TC-08	REQ-10	Edit attendance after 2 days and after 4 days	Record exists	2 days - saved; 4 days - blocked	High	Pass
TC-09	REQ-11	Check % after 3 of 4 classes attended	Records exist	75.00% shown	High	Pass
TC-10	REQ-12, NFR-8	Student opens own attendance; then changes URL to another student	Logged in as student	Own data shown; other student - 403	High	Pass
TC-11	REQ-13, REQ-14	Generate report for a date range in PDF and Excel	Attendance data present	Both files download with correct totals	High	Pass

TC-12	REQ-15	Shortage list with students at 74.99%, 75%, 80%	Threshold 75%	Only 74.99% student listed	High	Pass
TC-13	REQ-16	Student attendance drops below 75%	Student email set	Shortage email sent to student	Medium	Pass
TC-14	REQ-17 to REQ-19	Apply leave, faculty approves	Student absent on that day	Status Approved; day marked Leave, % updated	Medium	Pass
TC-15	NFR-2	Submit attendance for 70 students	Test data loaded	Saved in less than 2 seconds	Medium	Pass
TC-16	NFR-6	Check password stored in database	User created	Hashed value (pbkdf2_sha256$...)	High	Pass
TC-17	NFR-9	Stay idle for 31 minutes	Logged in	Session expired, redirected to login	Medium	Pass
TC-18	REQ-3, NFR-7	Create admin with createsuperuser, open admin dashboard and /admin	-	Role is Admin, both pages open; faculty and students cannot open /admin	High	Pass
TC-19	REQ-4	Upload Excel rows with bad SRN, long name, bad email, 2 letter section, long department	Logged in as admin	Each bad row rejected with its own reason, good rows saved	Medium	Pass
TC-20	REQ-17	Apply leave with very short reason, more than 30 days, or overlapping dates	Logged in as student	Request rejected with a message, nothing saved	Medium	Pass
TC-21	REQ-15	Shortage list filtered to one class	Logged in as faculty	Only students below 75% in that class shown	Low	Pass

Traceability Matrix

Requirement	**Test Case(s)**
REQ-1	TC-01, TC-02

REQ-2	TC-03
REQ-3	TC-05, TC-18
REQ-4	TC-04, TC-19
REQ-5 to REQ-7	TC-05
REQ-8	TC-06
REQ-9	TC-07
REQ-10	TC-08
REQ-11	TC-09
REQ-12	TC-10
REQ-13, REQ-14	TC-11
REQ-15	TC-12, TC-21
REQ-16	TC-13
REQ-17 to REQ-19	TC-14, TC-20
NFR-2	TC-15
NFR-6	TC-16
NFR-7	TC-03
NFR-8	TC-10
NFR-9	TC-17
NFR-10, NFR-11	Section 5.1 checks

14. Test Metrics & Reporting

- % of test cases executed
- % passed / failed
- Number of defects per module
- Defects by severity

- Requirement coverage %

Test status is reported at the end of every sprint and a final test summary report is made at the end.

Test execution evidence (29-10-2026)

Item	Details
Command	cd sams; USE_SQLITE=1 python manage.py test
Result	121 tests run, 121 passed, 0 failed (main c1edd6d + tests added on 02-11-2026)
CI	GitHub Actions run 40624798363 on main - success (github.com/SOFTWAREENGINEERINGMINIPROJECT/SE_mini_project/actions/runs/40624798363)
TC-15	Executed 02-11-2026: save 70 students 0.25 s, page loads under 0.3 s, semester report 0.07 s; 100 concurrent users median 0.67 s. NFR-1 to NFR-3 met
TC-11	Pass - report and PDF / Excel export (SEM-16) both work
Reports	docs/Unit_Test_Report.pdf, docs/System_Validation_Report.pdf, docs/Code_Coverage_Report.pdf (96.5% statements), docs/Branch_Coverage_Report.pdf (93.8% branches)

15. Approvals

Role	Name	Signature / Date
QA Lead	Epari Subhransi	
Dev Lead	Deepak B	
Team Lead	Gyaan S Shetty	
Faculty Guide		